MILITARY SPECIFICATION

SLED, SUBMARINE PIPELINE, LAUNCHING,

AND

ASSEMBLY, TERMINAL, SUBMARINE PIPELINE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a submarine pipeline launching sled and submarine pipeline terminal assembly with fittings for 8-, 12-, 16-, and 20-inch submarine pipelines.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 3835

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13213E8574	- Sled, Submarine Pipeline, Launching, and Assembly, Terminal, Submarine Pipeline.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN PETROLEUM INSTITUTE (API)

Standard 1104 - Standard for Welding Pipelines and Related Facilities.

(Application for copies should be addressed to the American Petroleum Institute, 2101 L Street, NW, Washington, DC 20037.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The submarine pipeline launching sled and submarine pipeline terminal assembly (hereinafter called "sled and assembly") shall be in accordance with Top Assembly TA13213E8574 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the sled and assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Identification marking. The sled and assembly shall be identified in accordance with MIL-STD-130.

3.4 Workmanship.

3.4.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.4.2 Welders and welding.

3.4.2.1 Pipeline seam welders. Before assigning any welder to manual welding of transverse pipe seams (i.e., welding the neck flange to the return bend), the contractor shall provide the contracting officer with certification that the welder has passed qualification tests in accordance with API 1104.

3.4.2.2 Structural steel welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of

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the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.4.2.3 Pipe seam welding. Transverse pipe seams (welding neck flange to return bend) shall be welded by the shielded metal-arc method in accordance with API 1104.

3.4.2.4 Structural steel welding. All structural steel welding, including anchor-riser grip assemblies and external welding on the submarine pipeline terminal assembly, shall be done by the shielded metal-arc method in accordance with AWS D1.1. The surface of all parts to be welded shall be free from paint, grease and other foreign matter. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part.

3.4.3 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads. Torque values shall be in accordance with the Appendix.

3.4.4 Machine work-anchor-riser grips. All components of the anchor-riser grip assemblies shall be machined to fit according to the tolerances shown on the drawings. The moving parts shall function without binding so that divers can easily assemble the anchor-riser grip underwater when a 1-inch-diameter wire rope is reeved through the riser slot.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.3 and 6.5.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 0.4 percent defective for major defects and 2.5 percent defective for minor defects.

4.4 Inspection procedure.

4.4.1 Examination. The sled and assembly shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the QAP requirements as shown in the drawings.
102. Material not as specified.
103. Parts or components missing or not as specified.
104. Poor fit at joints, seams or edges.
105. Welding defects such as flux inclusion, spatter, undercutting, porous burned metal, or lack of fusion with parent metal or between parts.
106. Assembly misalined.
107. Workmanship, except welding, not as specified.

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Minor

201. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
202. Identification marking missing, incomplete, incorrect, or not legible.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108. Materials or containers not as specified. Each incorrect material or container shall constitute one defect.
109. Disassembly not as specified for Level A.
110. Unpainted exterior metal surfaces requiring a preservative, not coated with preservative as specified.
111. Consolidated components not blocked, braced and cushioned for Level A.
112. Strapping not as specified for Level A.
113. Marking illegible, incorrect, incomplete or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The two anchor retainers and any other components subject to damage or loss shall be disassembled from the sled.

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5.1.1.2 Exterior surfaces. Unpainted exterior metal surfaces of components and attachments (such as bolts, nuts, washers, cables, keeper and wedge, threads, or any other unpainted exterior metal surfaces not specified herein) requiring the application of a contact preservative shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.1.1.3 Fasteners. The bolts, nuts and washers shall be preserved in a box conforming to PPP-B-636, W5c, style optional. The gross weight shall not exceed the limitations of the box specification. The box shall be closed in accordance with the box specification.

5.1.1.4 Consolidation. The preserved fasteners and disassembled components, except the anchor retainers, shall be consolidated in boxes conforming to PPP-B-601, Overseas Type. The contents shall be blocked, braced and cushioned to prevent damage.

5.1.1.5 Anchor retainers. The two anchor retainers shall be preserved together, with the bottoms of the retainers strapped together back-to-back, in a close-fitting box conforming to PPP-B-601, Overseas Type.

5.1.2 Commercial. The sled and assembly shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. The terminal assembly shall be secured in a flat position on the sled with steel strapping. The boxed anchor retainers and consolidation containers shall be placed on the sled and secured with steel strapping. Steel strapping shall conform to QQ-S-781, Type I, Finish B. Further packing of the sled is not required.

5.2.2 Commercial. Each complete sled and assembly shall be packed in accordance with MIL-STD-1188, except further packing of the sled is not required.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

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5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The sled and assembly are intended for use as part of an offshore submarine pipeline. The sled and assembly provide a means whereby the unloading hose and explosive embedment anchors can be installed onshore prior to launching of the submarine pipeline, thereby reducing the diving effort that would otherwise be required if they were installed after the pipeline had been pulled into place.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of this specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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APPENDIX

10. SCOPE

10.1 Scope. This appendix details the torque data for nuts and screws used with the sled and assembly.

Nut or screw torque data

Size Nut or Screw	Torque (Ft.-Lb.)
1/4 - 20	7-9
1/4 - 28	8-10
5/16 - 18	13-17
5/16 - 24	15-19
3/8 - 16	30-35
3/8 - 24	35-39
7/16 - 14	46-50
7/16 - 20	57-61
1/2 - 13	71-75
1/2 - 20	83-93
9/16 - 12	90-100
9/16 - 18	107-117
5/8 - 11	137-147
5/8 - 18	168-178
3/4 - 10	240-250
3/4 - 16	290-300
7/8 - 9	410-420
7/8 - 14	475-485
1 - 8	580-590
1 - 14	685-695

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-S-52551B(ME) Sled, Submarine Pipeline, Launching, and Assembly, Terminal, Submarine Pipeline

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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